RIP asoslari

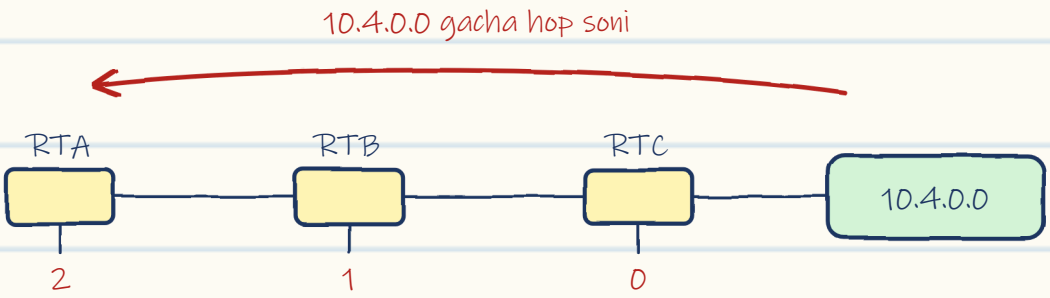
RIP Fundamentals

Asosiy atamalar
- RIP — eng sodda masofa-vektor protokoli, UDP 520
- Metrika (hop) — manzilgacha nechta marshrutizator borligi
- 16 = cheksizlik — yaroqli qiymat 0–15, 16 — yetib bo'lmaydi
- Taymerlar — update 30 s, timeout 180 s, garbage-collect 120 s
- Split horizon — marshrut o'rganilgan portga qaytarib e'lon qilinmaydi
- Poison reverse — qaytariladi, lekin metrika 16 bilan

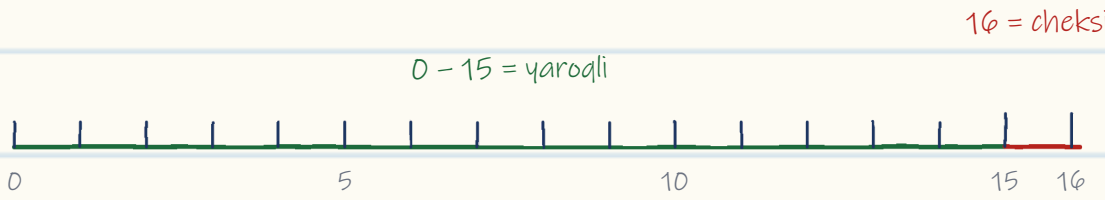
1. RIP nima?

Distance-vector IGP — masofa hop soni bilan o'lchanadi (Routing Information Protocol)

- metrika 0...15 — yaroqli, 16 = cheksiz (erishilmas)
- UDP 520 — xabarlar shu portda almashinadi
- kichik va o'rta tarmoqlar; OSPF va IS-IS dan sodda
- ikki versiya: RIPv1 (klassli) va RIPv2 (klasssiz)



1-rasm. Metrika — manzilgacha bo'lgan hop soni



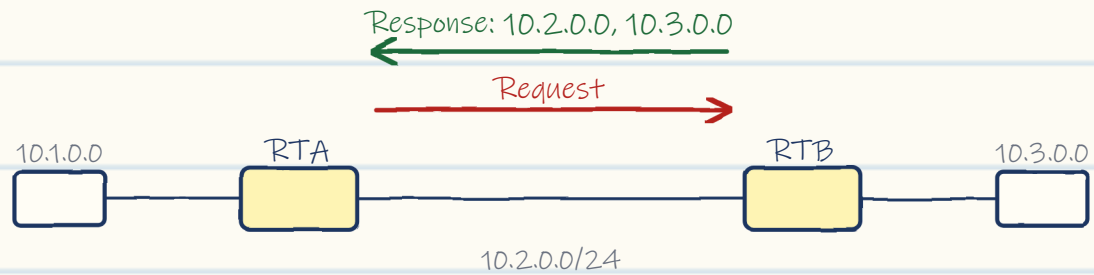
16 va undan katta — erishilmas

shuning uchun katta tarmoqqa mos emas

2-rasm. Metrika diapazoni va konvergensiya chegarasi

2. Jadval qanday quriladi

- boshda jadvalda faqat bevosita marshrutlar
- Request yuboriladi → qo‘shnilar Response bilan jadvalini beradi



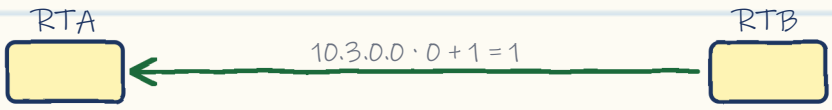
RTA jadvali — yangilangach

Tarmoq	Keyingi uzel	Metrika
10.1.0.0	—	0
10.3.0.0	10.2.0.2	1

3-rasm. 10.2.0.0 bevosita bo‘lgani uchun jadvalga qo‘shilmaydi

3. Yangilash qoidasi

- o‘sha qo‘shnidan kelsa — metrika katta ham, kichik ham bo‘lsa yangilanadi
- boshqa qo‘shnidan — faqat metrika kichik bo‘lsa
- jadvalda yo‘q marshrut — metrika 16 dan kichik bo‘lsa qo‘shiladi

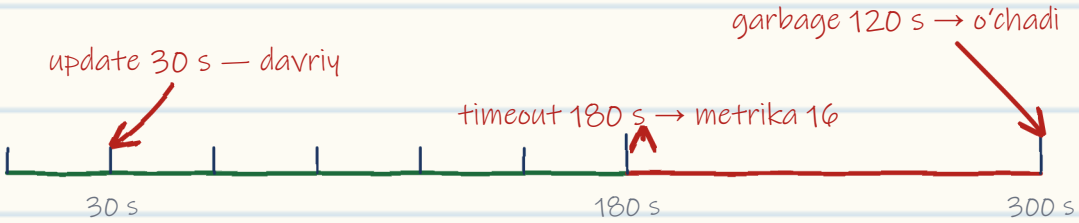


chiqishda +1 · kirishda +0

RTA jadvaliga metrika 1 bilan tushadi

4-rasm. Chiqish metrikasi sukut bo'yicha 1, kirish metrikasi 0

4. Taymerlar



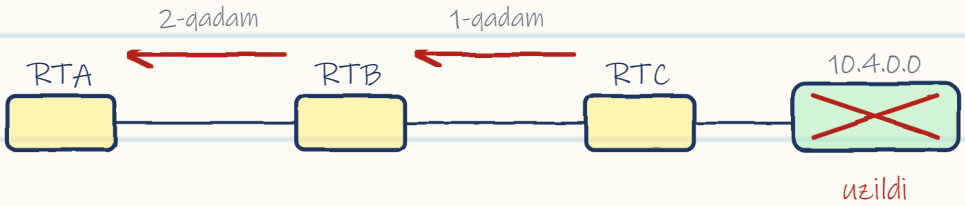
suppress = 120 s · yaroqli marshrut IP jadvaliga tushadi

yangilanish kelmasa — 16, keyin o'chiriladi

5-rasm. Marshrutning umri: 30 s → 180 s → yana 120 s

update	30 s	timeout	180 s
garbage-collect	120 s	suppress	120 s

5. O'zgarish hop-by-hop tarqaladi

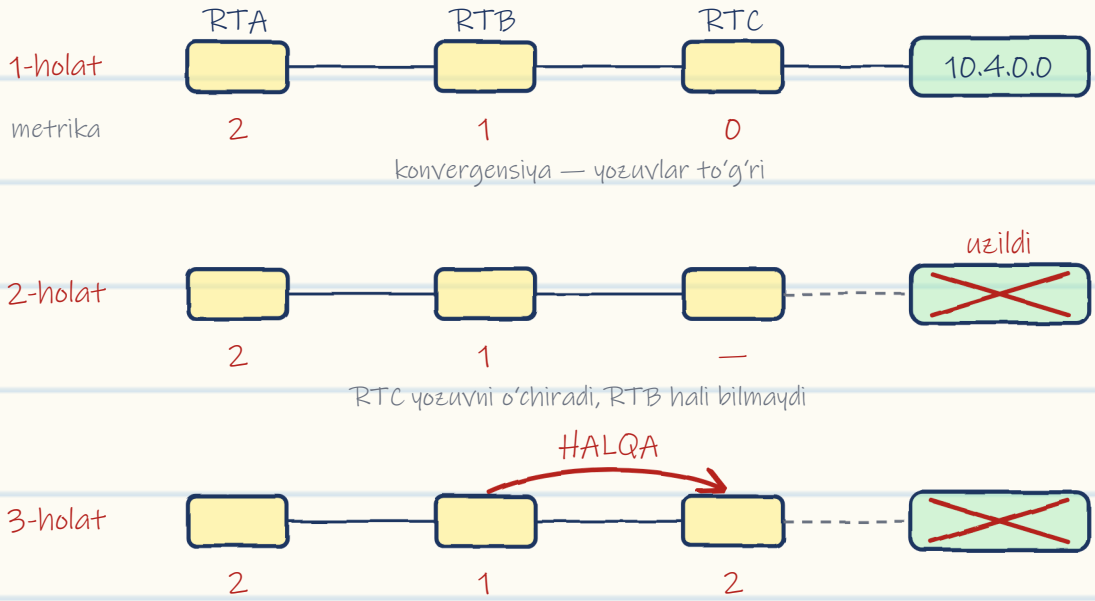


har router faqat qo'shnisiga aytadi

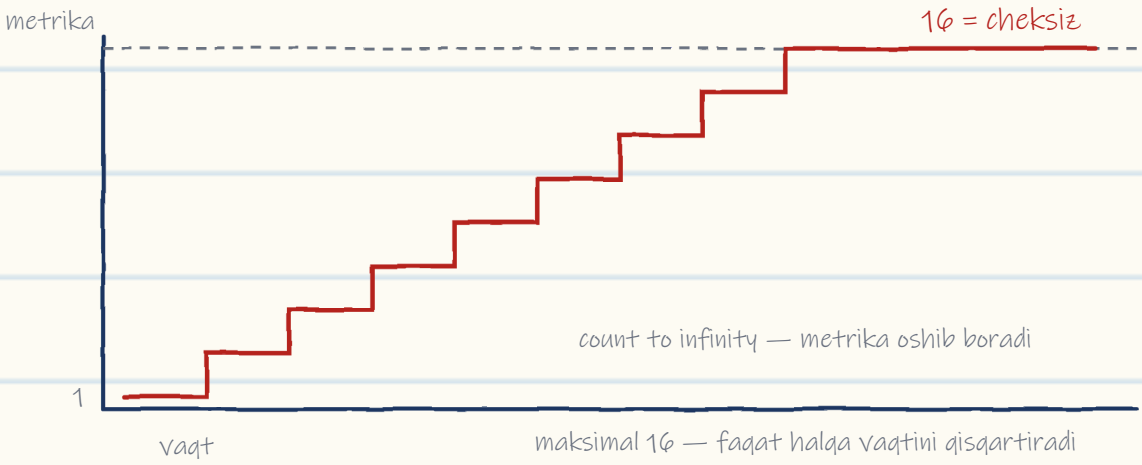
tarqalish vaqti = router soni · update timer

6-rasm. Router marshrut manbasini bilmaydi — faqat qo'shnisidan oladi

6. Halqa qanday tug‘iladi



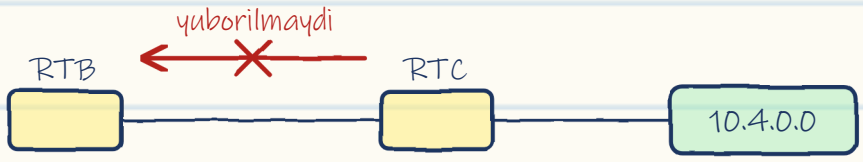
7-rasm. RTB uchun yo‘l RTC orqali, RTC uchun RTB orqali



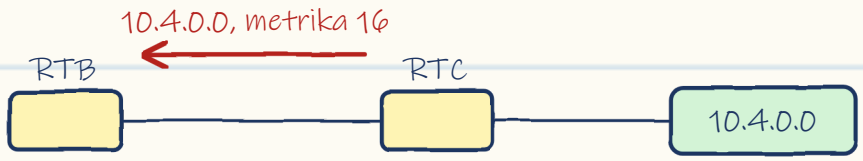
8-rasm. Ko‘p yo‘lli tarmoqda metrika 16 ga yetguncha o‘sadi

- sabab — router marshrut manbasini bilmaydi

7. Halqadan saqlanish



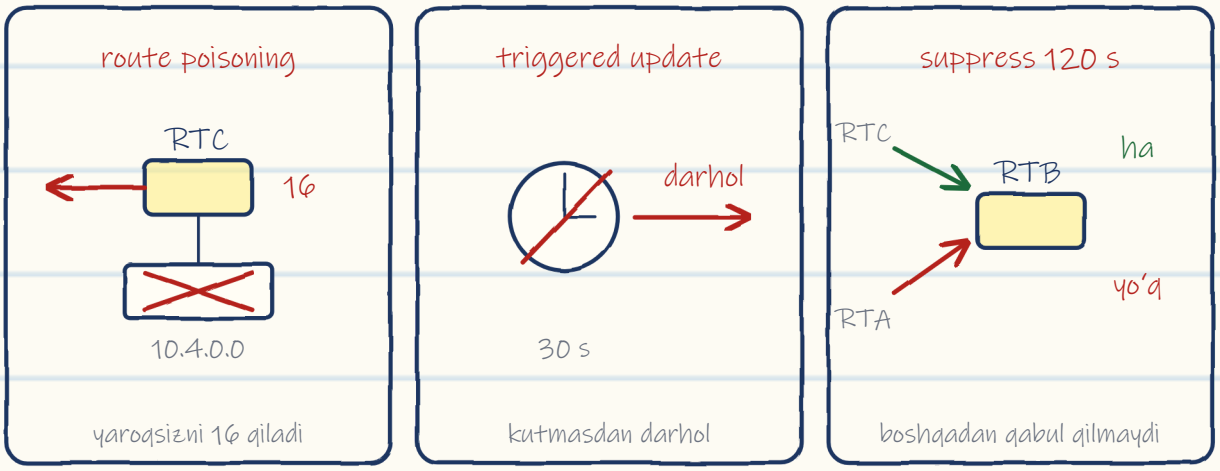
split horizon: S1/0 dan olingan marshrut S1/0 ga qaytmaydi



poison reverse: 16 bilan orqaga qaytariladi — ochiq rad

+ ishonchli · – trafik ko'p

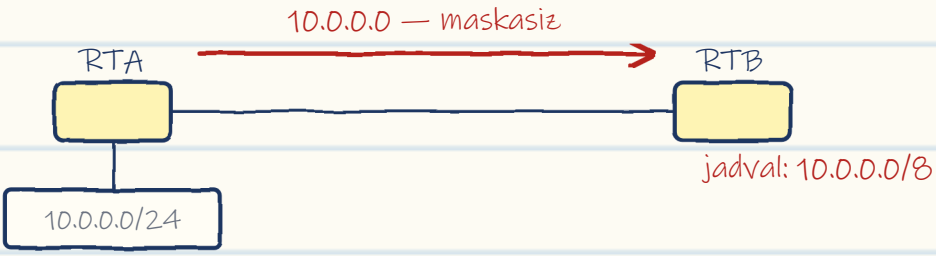
9-rasm. Split horizon — jim turadi, poison reverse — ochiq aytadi



10-rasm. Suppress paytida faqat o'sha qo'shnidan 16 dan kichik metrika olinadi

Amalda: hammasi birga ishlatiladi — poisoning + suppress + triggered update

8. RIPv1 kamchiligi



RIPv1 klassli — tabiiy maska qo'yiladi → xato marshrut

11-rasm. Maska yo'qligi subnet adreslashda xatoga olib keladi

	RIPv1	RIPv2
Maska	yo'q — klassli	bor — VLSM, CIDR
Tarqatish	faqat broadcast	multicast 224.0.0.9
Autentifikatsiya	yo'q	oddiy matn, MD5

Xulosa

- RIP — distance-vector, metrika = hop, 0...15, 16 = cheksiz
- UDP 520 · update 30 s · timeout 180 s · garbage 120 s
- Marshrut hop-by-hop tarqaladi — manba bilinmaydi
- Halqa sababi: eski ma'lumot davriy yangilanishda qaytadi
- Yechim: split horizon · poisoning · poison reverse · suppress
- RIPv2: VLSM/CIDR, multicast 224.0.0.9, autentifikatsiya

Tekshirish

1. Qaysi port? — UDP 520
2. Cheksiz metrika? — 16
3. Timeout va garbage? — 180 s va 120 s
4. Split horizon? — olgan interfeysga qaytarmaydi
5. RIPv2 multicast adresi? — 224.0.0.9